PORTFOLIO MANAGEMENT & QUANTITATIVE TRADING

Deep Sequence Models in Commodity Markets

Testing and Optimizing the LSTM Momentum Hypothesis within the skfin Backtesting Engine

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01 / RESEARCH HYPOTHESIS

Can Regularized LSTMs Beat Linear Momentum?

CORE OBJECTIVE

Net Sharpe Optimization After Market Impact

Evaluating compact recurrent networks against Simple Momentum, RidgeCV, and LightGBM across five liquid commodity ETFs.

Time-Series Momentum: Multi-horizon trend continuation across 1-week to 12-month lookbacks.

Deep Sequence Learning: Nonlinear regime extraction under strict capacity constraints.

Quadratic Cost Policy: Trading partially toward Markowitz aim portfolios.

Asset Pricing Hygiene: Leak-free walk-forward splits with multi-seed ensembling.

Cross-Sector Volatility Dispersion & Fat Tails

Ticker	Commodity Sector	Ann. Mean	Ann. Vol	Passive SR	Kurtosis	Max Crash
DBC	Broad Diversified	5.92%	19.36%	0.31	3.46	-7.94%
USO	Crude Oil / Energy	2.53%	**37.49%**	**0.07**	9.47	**-25.32%**
GLD	Gold Bullion	10.87%	18.14%	**0.60**	7.17	-10.27%
SLV	Silver Bullion	12.73%	33.00%	0.39	**13.41**	-28.54%
DBA	Agriculture Basket	3.94%	16.68%	0.24	6.41	-8.61%

Portfolio Implication: 2x volatility spread and low cross-sector correlation (GLD–USO = 0.20) necessitate covariance scaling.

Point-in-Time Walk-Forward Architecture

STEP 01

Data Alignment

5,158 business days synchronized from DBA inception.

STEP 02

25-Feature Tensor

5d, 21d, 63d, 126d, 252d rolling momentum scales.

STEP 03

Rolling Retraining

756-day training window updated every 21 days.

STEP 04

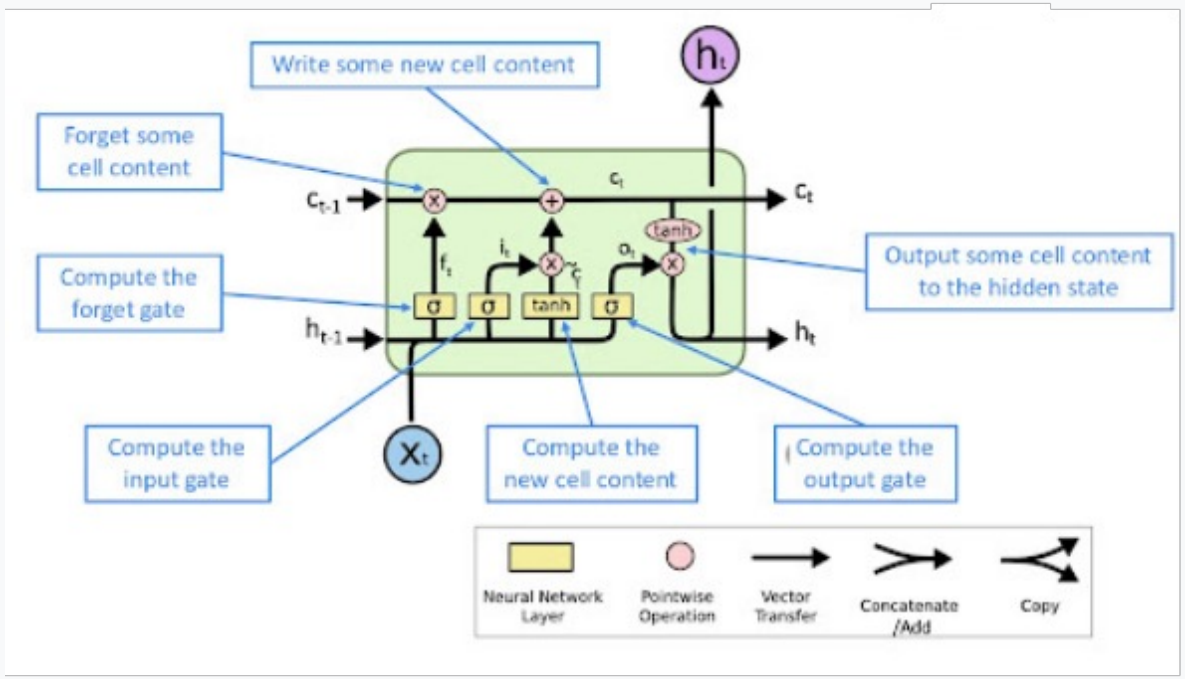
Risk Scaling

Markowitz weights scaled to 10% annual volatility.

Zero Look-Ahead Invariant: Next-day target $y(t) = r(t+1)$ evaluated via $h(t) \cdot r(t+1)$

TARGET_VOL = 0.10 / √260

Mapping Commodity Data to the LSTM Cell



Input Vector (xt): 25 standardized multi-horizon ETF momentum features daily.

Forget & Input Gates (ft, it): Prunes past noise and writes new trend signals.

Cell Memory (ct-1 → ct): Propagates 20-day regime context across time steps.

Hidden State (ht): 8-unit bottleneck summary emitted via output gate.

Return Readout (hT → ŷt): Linear head maps terminal hT to 5 returns.

05 / DEEP SEQUENCE SAFEGUARDS

Anti-Overfitting Safeguards for Noisy Returns

Compact Bottleneck (hs = 8)

Eight hidden units prevent in-sample return memorization.

Tail-Robust Huber Loss (δ = 0.01)

Linear tail penalty dampens extreme oil crash outliers.

Dropout & L2 Decay (1e-3)

Strong weight shrinkage prunes spurious recurrent parameters.

Walk-Forward Buffer (_history_X)

Cached trailing training rows preserve boundary sequence continuity.

Why Pre-2020 Hyperparameters Fail in Crisis Regimes

2012–2019 CALIBRATION WINNER (TRIAL 9)

Long-Memory LSTM (lookback = 32)

Parameters: 9 units, 32d window, weak decay (3.4e-4).

2012–2019 Sharpe: 0.65 in tranquil bear market.

2020–2026 OOS Sharpe: Collapses to -0.07 via memory lag.

CRISIS-ROBUST SPECIFICATION

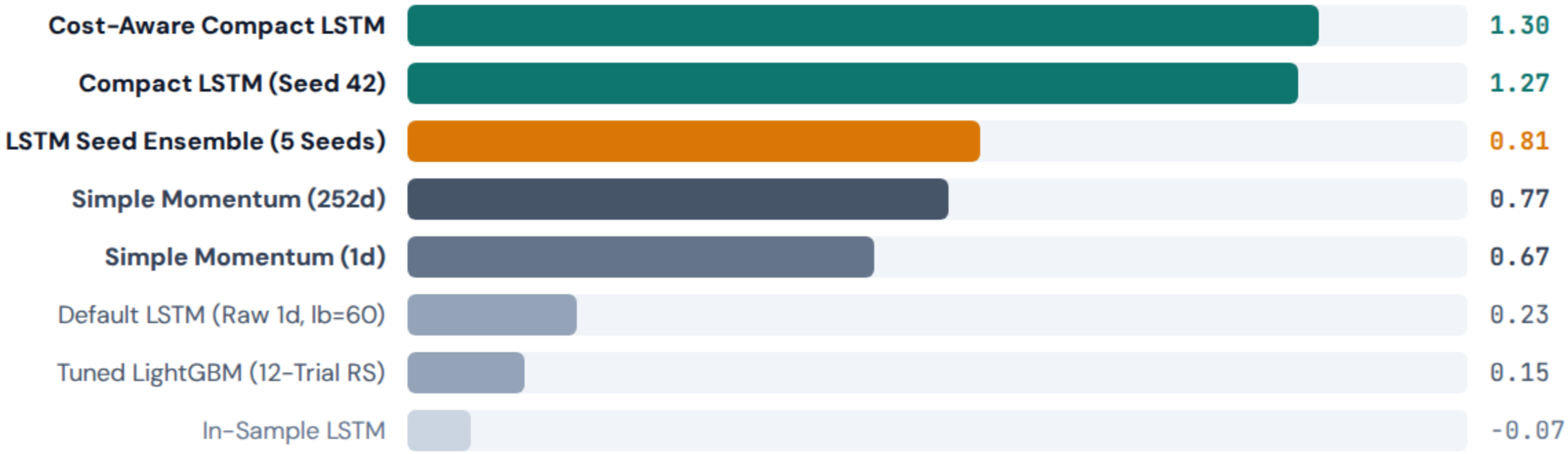
Compact Regularized LSTM (lookback = 20)

Parameters: 8 units, 20d window, strong decay (1e-3).

Turning-Point Agility: Flushes pre-shock states within 1 month.

2020–2026 OOS Sharpe: Surges to 1.27 (0.81 Ensemble).

Walk-Forward Gross Sharpe Ratios (10% Target Volatility)



Decomposing Feature Synergy & Seed Ensembling

1.27 vs 0.20

2×2 Ablation Synergy

Multi-horizon inputs require compact 20-day memory.

0.81 SR

Five-Seed Ensemble

Equal-weighted holdings eliminate initialization variance.

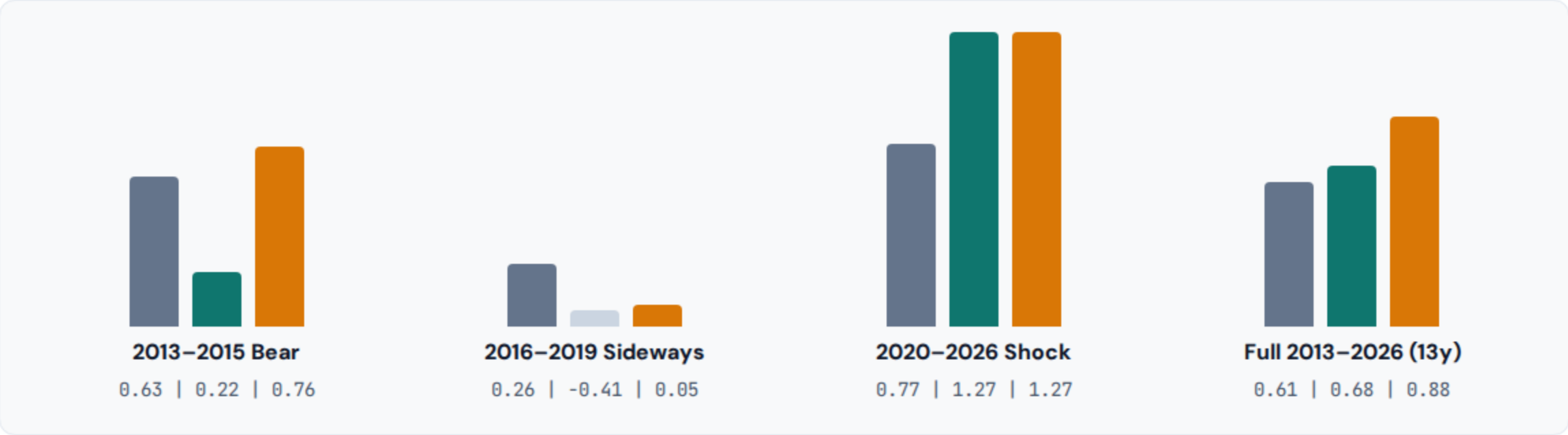
-49% Turn

Turnover Smoothing

Ensembling cuts annual trading volume in half.

09 / HISTORICAL SUBPERIODS & WINDOW SENSITIVITY (2013–2026)

3-Year Rolling Window Balances Variance & Stationarity



■ Simple Mom 252d (3y Win)

■ Compact LSTM (lb=20, 3y Win)

■ Regime-Adaptive LSTM

Window Trade-off (2013–2026): 3y (0.68) > 2y (0.35) > 5y (0.05)

Net Performance After Volatility-Adjusted Transaction Costs

Strategy	Gross Ret	Net Ret	Cost Drag	Gross SR	Net SR	Max DD	Avg Hold
Cost-Aware Compact LSTM	**15.94%**	**15.58%**	**-0.36%**	**1.30**	**1.26**	**-16.68%**	**8.8d**
Compact LSTM (Seed 42)	17.56%	16.71%	-0.84%	1.27	**1.21**	-22.18%	9.4d
LSTM Seed Ensemble (5 Seeds)	4.78%	4.60%	**-0.18%**	0.81	**0.78**	**-5.89%**	6.0d
Simple Momentum (252d)	10.70%	10.52%	-0.18%	0.77	0.76	-17.58%	33.0d
Simple Momentum (1d)	10.18%	4.04%	**-6.14%**	0.65	0.26	-48.22%	1.5d
Tuned LightGBM (RS)	2.23%	-0.94%	-3.16%	0.15	-0.06	-41.43%	3.3d

Empirical Summary of the LSTM Momentum Edge

1.26

Cost-Aware Net Sharpe

Outperforms 252d Momentum (0.76)

-0.36%

Annual Cost Drag

57% lower market impact via past_h

0.88

13-Year Adaptive SR

Full 2013–2026 multi-regime span

-5.89%

Ensemble Max Drawdown

Versus -17.58% for 252d Momentum

HAC Inference (Lo, 2002): Autocorrelation-adjusted standard errors confirm positive out-of-sample alpha.

Universe Stability: Positive net alpha persists under leave-one-ETF-out tests.

CONCLUSION & RESEARCH VERDICT

Compact Multi-Horizon LSTMs Outperform After Transaction Costs

Pairing 25 multi-scale momentum features with a 20-day Huber-regularized LSTM and cost-aware Markowitz optimization delivers superior risk-adjusted commodity returns.

Thank You — Questions & Discussion